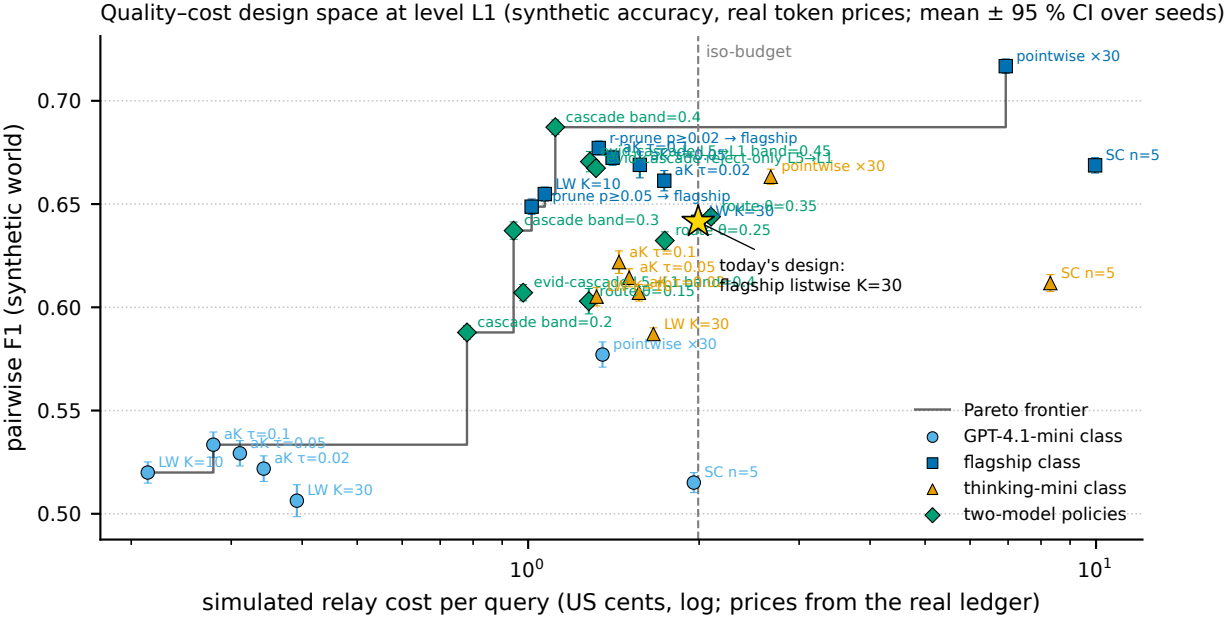
Quality–cost design space at level L1 (synthetic accuracy, real token prices; mean ± 95 % CI over seeds)
iso-budget
pointwise ×30
SC n=5
cascade band=0.4
r-prune p≥0.02 → flagship
aK τ=0.02
LW K=10
prune p≥0.05 → flagship
LW K=30
cascade band=0.3
today's design:
flagship listwise K=30
aK τ=0.1
aK τ=0.05
SC n=5
LW K=30
cascade band=0.2
pointwise ×30
aK τ=0.1
aK τ=0.05
aK τ=0.02
LW K=10
LW K=30
SC n=5
Pareto frontier
GPT-4.1-mini class
flagship class
thinking-mini class
two-model policies
pairwise F1 (synthetic world)
0.70
0.65
0.60
0.55
0.50
10^0
10^1
simulated relay cost per query (US cents, log; prices from the real ledger)